Horizon at a glance: what each part is for, and what RQ1 does not need

RQ1 asks about INDIRECT prompt injection. Grey crossed-out parts target other attacks or only duplicate a test.

Attacks we test with

Adaptive attackers
an attacker LLM rewrites δ until it works
● indirect

BIPIA · LLMail-Inject
attack hidden in a document
● indirect

AgentDojo injections
attack hidden in emails, files, web pages
● indirect

Guard detection sets
HackAPrompt · HarmBench · JBB · Gandalf · XSTest
× not needed: tests the guards only
● direct ● jailbreak

garak probes
× not needed: tests the bare model
● direct ● jailbreak

PyRIT XPIA
× optional: repeats the AgentDojo test
● indirect

PyRIT PAIR · TAP · Crescendo
× not needed: jailbreaks, not RQ1
● jailbreak

documents

poisoned tool data

Stage 1 · Keyed spotlighting

the reader alone: one document, a fixed question, no user agent
● indirect

Mark the document
random marking τ, secret key κ

Reader LLM
answers; canary shows if it obeyed δ

Clopper–Pearson bound

▼ **Adaptive stress test**

Forging bound

▼ **DRO (χ^2)**

Choice rule (utility constraint)

▼ **Combined risk (max)**

McNemar · bootstrap
support statistics

▼ **Sion minimax game, level 1**
→ π*: which markings, how often

π*

Stage 2 · Deterministic guardrail (agent with tools)

one user task, AgentDojo tools, the attack hides in the data the tools return

Input guard (DeBERTa)
× not needed: checks what the USER types
● direct ● jailbreak

Planner (P-LLM)
never sees untrusted data
● indirect

Interpreter + data labels
● indirect

Quarantined readers × m
Stage 1 marking inside every read
● indirect

Forging bound (per read)

Typed schema + majority vote
● indirect

Leakage bound (QIF) · Hoeffding vote bound

Policy table
ALLOW · CONFIRM · DENY
● indirect

Noninterference proof (Z3)

Ask the user or deny
threshold $\lambda^\wedge$ on reader agreement
● indirect

Conformal risk control

Tool call

Output guard (Qwen3Guard) + canary
× not needed: judges harmful answers
● jailbreak ● direct

Answer

final, or through

Horizon value · both stages in one bound

● indirect

▼ **DRO on the loss**

▼ **Weighted loss per attack**
policy bypass (heaviest) · reached the user · fooled the vote

▼ **Sion minimax game, level 2 → V_H**
under utility and leakage budgets

Test setups

Model pairing · autotune
× one-off choices, not part of the pipeline

Stage 1 only · Stage 2 only · Stage 1+2
× optional: only if each stage's share is asked

No defence vs full Horizon
the comparison RQ1 needs · + calibration for $\lambda^\wedge$

● **indirect**: attack hidden in data the model reads
● **direct**: attack typed by the user
● **jailbreak**: talk the model into harmful output
▼ **thick black border**: worst case weighs most ("black hole")
green: math concept · **lavender**: LLM
grey, dashed, ×: not needed for RQ1